

Name: _____

Double Award Science – Chemistry Unit 2 Higher Tier

Once you have completed a paper and have marked it, fill out the table below, using the grade boundaries provided to add your grade. Then list which questions you did well and which questions you struggled with. For the questions you struggled with, review why and at the reasons. The most common reasons are

- Not understanding basic concepts (e.g. simple definitions)
- Calculation errors (forgetting to square numbers, etc.)
- Not reading questions carefully
- Not answering the full question (describe vs. explain)
- Laboratory/practical setup mistakes

For these questions it is also useful add notes to the question to help you next time and take a photo of these questions. Use these then to guide further revision.

Year	Score /60	Grade	Good Questions	Bad Questions	Reasons
2017					
2018					
2019					
2022					
2023					

Grade boundaries - Raw Score

Year	E	D	C	B	A	A*	Max
2017	6	9	15	21	28	35	42
2018	4	7	13	21	30	39	48
2019	4	8	13	19	25	31	37
2022	4	6	13	20	27	34	41
2023	5	7	11	18	26	34	42

Surname	Centre Number	Candidate Number
Other Names		0



GCSE – **NEW**

3430UE0-1



SCIENCE (Double Award)

**Unit 5 – CHEMISTRY 2
HIGHER TIER**

THURSDAY, 17 MAY 2018 – MORNING

1 hour 15 minutes

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	11	
2.	4	
3.	11	
4.	12	
5.	8	
6.	8	
7.	6	
Total	60	

ADDITIONAL MATERIALS

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen. Do not use correction fluid.
Write your name, centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.
Question **7** is a quality of extended response (QER) question where your writing skills will be assessed.
The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

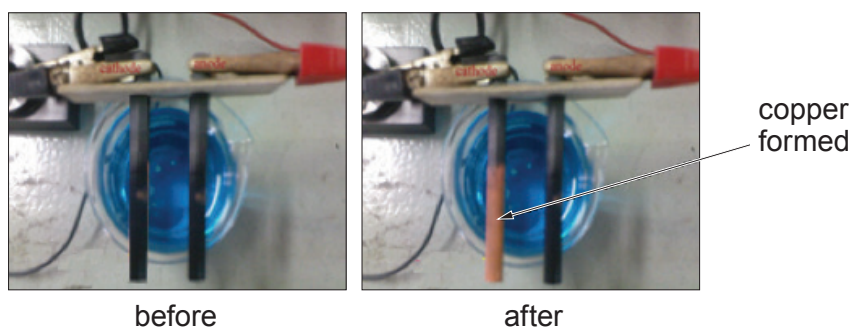
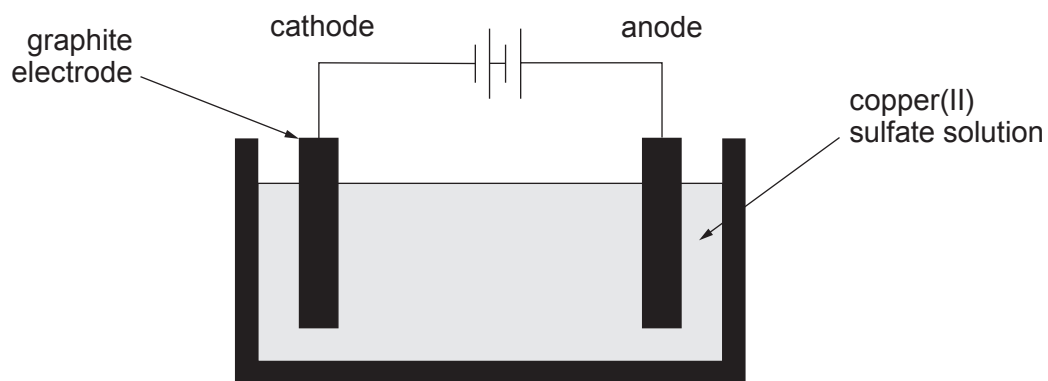


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Answer **all** questions.

1. A group of students carried out an investigation into the electrolysis of copper(II) sulfate solution. They used the apparatus shown to test the hypothesis:

“the mass of copper that forms on the cathode increases as the time increases”



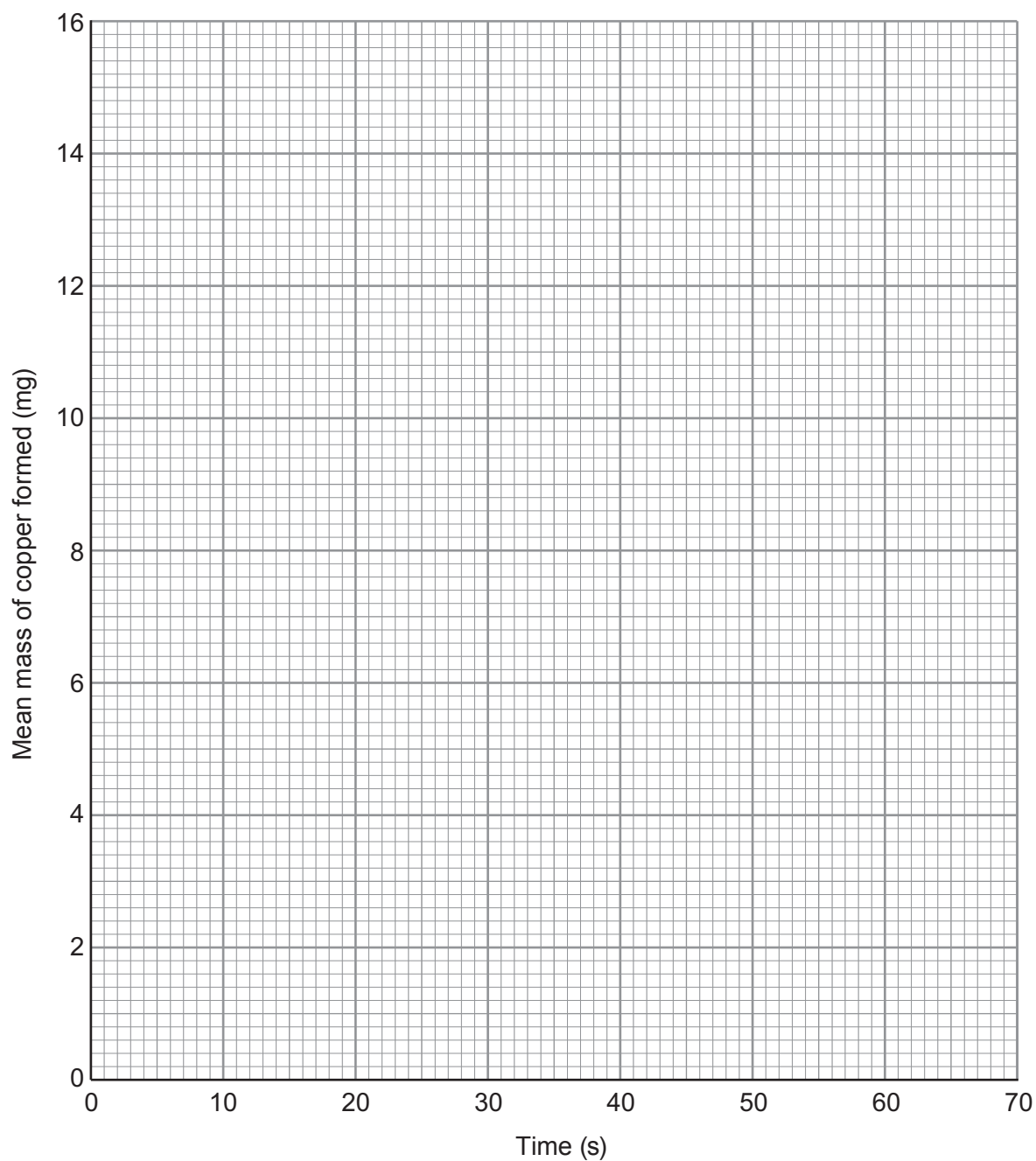
To test the hypothesis, they weighed the cathode before placing it into the copper(II) sulfate solution and then again after allowing electrolysis to take place for varying times.

Their results are shown below.

Time (s)	Mass of copper formed (mg)		
	1	2	Mean
0	0	0	0
10	2.8	3.2	3.0
20	4.8	5.0	4.9
30	8.2	7.8	8.0
40	10.8	11.2	11.0
50	12.9	13.1	13.0
60	15.8	16.0	15.9



- (a) On the grid below, plot the mean mass of copper formed against time. Draw a suitable line. [3]



- (b) (i) Use the results collected at 30s and the following equation to calculate the percentage variation in these measurements. [2]

$$\text{percentage variation} = \frac{\text{furthest mass from the mean} - \text{mean mass}}{\text{mean mass}} \times 100$$

Percentage variation = %

- (ii) The mass of copper formed is lower than expected. Give the most likely reason for this difference. [1]

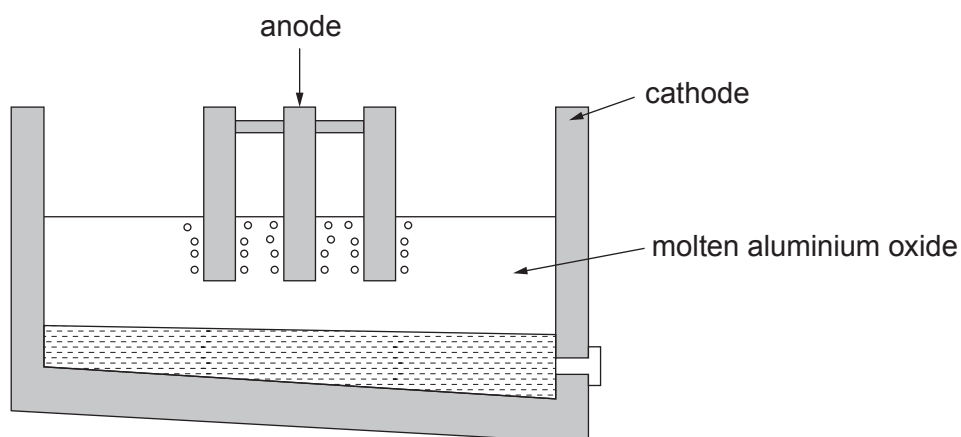
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- (c) (i) Aluminium is extracted from molten aluminium oxide by electrolysis.



- I. Explain why aluminium forms at the cathode.

[2]

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- II. Complete and balance the equation for the overall reaction that takes place.

[2]



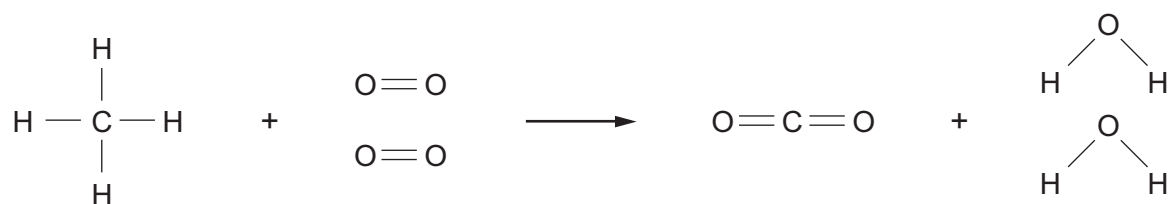
- (ii) Potassium can also be extracted through electrolysis of potassium carbonate.

Write the **formula** of potassium carbonate to complete the equation for the overall reaction.

[1]



2. The burning of methane in air can be represented by the following equation.



The bond energies are given in the table below.

Bond	Bond energy (kJ)
C — H	413
O = O	498
O — H	464
C = O	805

- (a) Use the bond energy values to calculate the energy released when **all** the bonds in the carbon dioxide and water molecules are formed. [2]

Energy released = kJ



- (b) The energy needed to break **all** the bonds in the methane and oxygen molecules is 2648 kJ.

Calculate the overall energy change for this reaction and use this value to explain why the reaction is exothermic. [2]

Overall energy change = kJ

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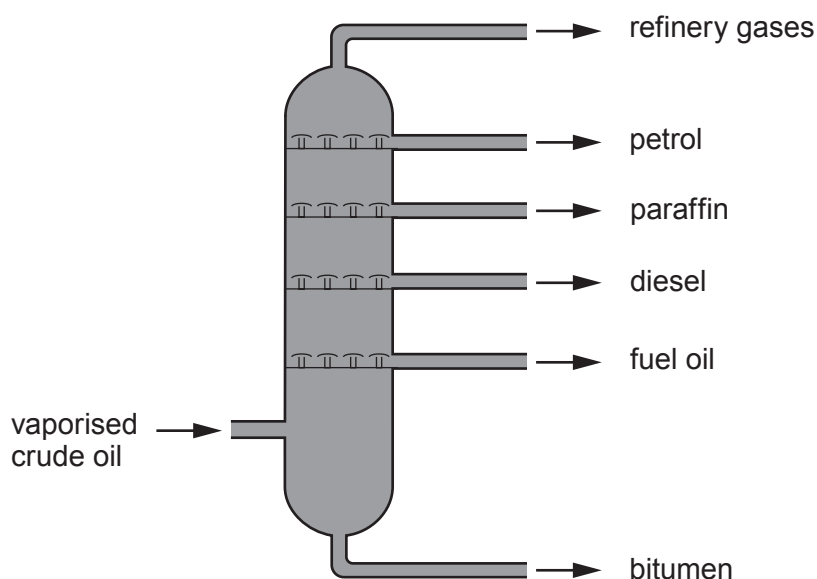
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3. Crude oil is a mixture of hydrocarbon compounds. It is separated into different fractions inside a fractionating column. Each fraction has a different boiling point range.



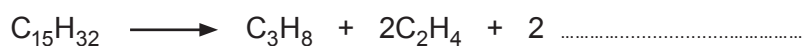
- (a) Explain how the length of the hydrocarbon chains within each fraction determines where each fraction collects in the column. [2]

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- (b) Many of the hydrocarbons that are collected from crude oil go through a second process called cracking. The equation shows the cracking of the hydrocarbon with the formula $C_{15}H_{32}$.



- (i) **Complete the equation** for this reaction by adding the **formula** of **one** other product. [1]

- (ii) Give the reaction conditions used in the process. [1]

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- (iii) Give **one** reason why it is important for oil companies to carry out cracking. [1]

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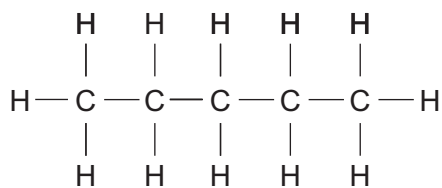


(c) Many hydrocarbons display isomerism.

(i) Give the meaning of the term *isomer*.

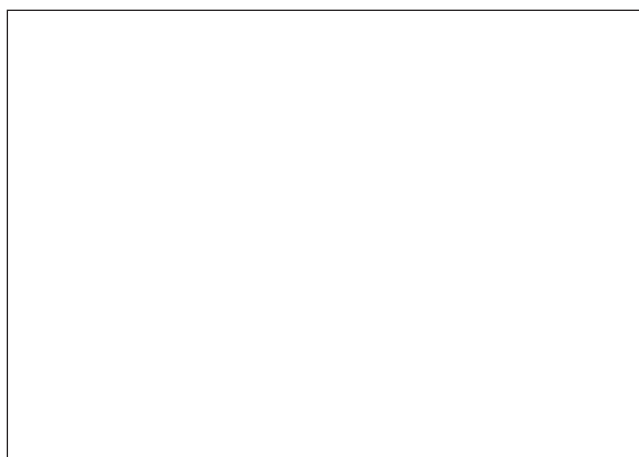
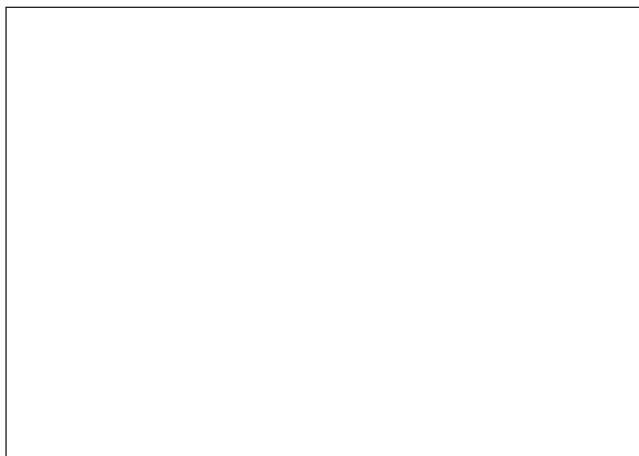
[1]

(ii) C_5H_{12} has three isomers. The diagram shows one of these isomers.



Draw the **two** other isomers of C_5H_{12} .

[2]



- (d) 8.4 g of a hydrocarbon was found to contain 7.2 g of carbon. Use this information to determine whether the compound is an alkane or an alkene. You **must** show your working. [3]

$$A_r(\text{H}) = 1 \quad A_r(\text{C}) = 12$$

Conclusion

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Examiner
only

11



4. Potassium chloride and potassium sulfate are salts that can be made from the reactions of acids. Becky and David carried out experiments to prepare both salts.

- (a) To make a sample of potassium chloride, they reacted potassium carbonate powder with dilute hydrochloric acid, HCl. They used the following method.

Stage 1 – add potassium carbonate powder to dilute hydrochloric acid

Stage 2 – filter the mixture

Stage 3 – leave the remaining mixture in a warm place overnight

- (i) David said it was important to know the volume of acid in stage 1.

Circle whether you agree or disagree with his statement and explain your answer. [2]

Agree / Disagree

Explanation

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.....

- (ii) Give the balanced **symbol** equation for the reaction taking place in stage 1. [3]

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- (b) To make a sample of potassium sulfate, they reacted potassium hydroxide solution with dilute sulfuric acid, H_2SO_4 . They used the following method.

Stage 1 – measure out 25.0 cm^3 of potassium hydroxide solution into a flask

Stage 2 – add 4-5 drops of indicator

Stage 3 – record the volume of sulfuric acid required to neutralise the solution

Stage 4 – repeat without the indicator, using the volume of acid used in stage 3

Stage 5 – leave the remaining mixture in a warm place overnight

- (i) Becky was concerned that carrying out stages 1-3 only once would mean that the volume of sulfuric acid used in stage 4 would be larger than that actually needed to neutralise the potassium hydroxide solution. Suggest the reason for her concern.

[1]

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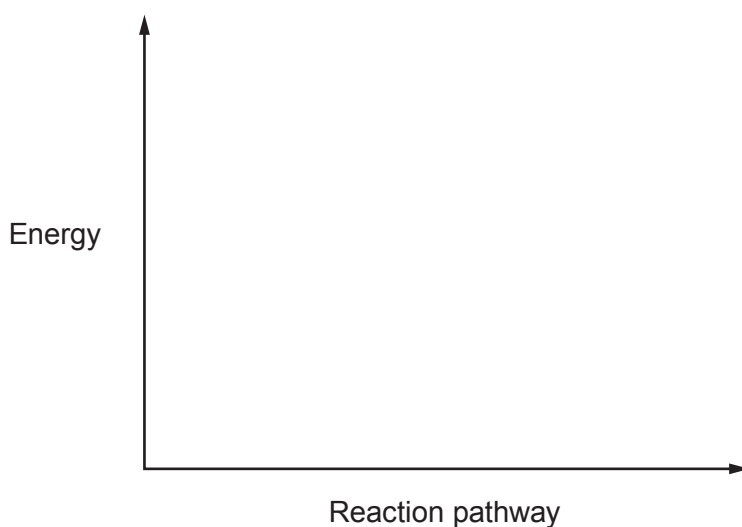
- (ii) Write the **ionic** equation to show how water is formed during this reaction and state the source of each of the ions.

[2]

Equation

Source of ions

-
- (iii) The reaction is exothermic. Sketch the energy profile diagram for this reaction. [1]



(c) Becky and David carried out a series of tests to identify the ions present in their salts.

- (i) They carried out a flame test on each of the salts. Give the reason why the results of this test could not be used to identify the salts. [1]

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- (ii) Describe a test that enabled them to tell the salts apart. Give the results seen with both salts. [2]

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5. (a) A teacher showed her class how copper can be extracted from copper(II) oxide using charcoal.



In her experiment, the teacher heated 1.59 g of copper(II) oxide and 0.12 g of charcoal in a boiling tube. She continued to heat the mixture strongly until it had been glowing for 5 minutes.

She recorded the mass of the boiling tube and its contents before heating and then again after heating, once the carbon dioxide produced had been released from the tube.

Mass of boiling tube = 37.43 g

Mass of boiling tube and contents **before** heating = 39.14 g

Mass of boiling tube and contents **after** heating = 38.82 g

- (i) The teacher had expected the reaction to produce 1.27 g of copper from the masses of copper(II) oxide and charcoal used. Use her results to show that the mass produced suggests a 109 % yield. [2]

- (ii) Assuming the reactants had been heated at a high enough temperature for sufficient time, suggest **one** reason for the yield being larger than expected. [1]

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(b) Iron is extracted from iron(III) oxide, Fe_2O_3 , inside the blast furnace using coke.

- (i) Write a balanced **symbol** equation for **one** of the reactions that show the reduction of iron(III) oxide inside the furnace. [2]

-
- (ii) An iron ore contains 22% by mass of iron(III) oxide. Calculate the maximum mass of iron that could be obtained from 5×10^5 tonnes of this ore. Give your answer in **standard form**. [3]

Mass = tonnes

8



6. Saturated and Unsaturated Fats

Saturated and unsaturated fats are found in different amounts in different foods.

Unsaturated fats are often described as 'good fats'. When used to replace saturated fats in the diet they help lower cholesterol, one of the risk factors linked to heart disease.

While full-fat dairy products and meat contain large amounts of saturated fats, sources of unsaturated fats include nuts, seeds and vegetable oils.

Saturated and unsaturated fats differ in their chemical structures. Unsaturated fats contain double bonds between the carbon atoms within their molecular structure, whereas saturated fats have no double bonds.



The label on the bottle of a new brand of cooking oil claims that it is better than other cooking oils available on the market because it contains only **15% saturated fat**.

Scientists decided to investigate this claim.

They used bromine water to compare the percentage unsaturation of this new brand of cooking oil with four other vegetable oils, **W**, **X**, **Y** and **Z**. They also tested a known alkane and alkene.

They measured the volume of bromine water that could be added to each oil sample before the bromine water colour remained. Their results are shown below.

Sample	Mean volume of bromine water added (cm ³)	Unsaturation (%)	Saturation (%)
oil W	20.4	50	50
oil X	12.0	30	70
oil Y	24.1	60	40
oil Z	16.3	40	60
new brand oil	30.1	?	?
alkane	0.1	-	-
alkene	40.0	-	-



- (a) Which **one** of these statements best describes the oils tested? Tick (✓) the correct answer. [1]

the oils contain saturated fats only

☐

the oils contain unsaturated fats only

☐

the oils contain both saturated and unsaturated fats

☐

it is not possible to tell whether the oils contain saturated or unsaturated fats

☐

- (b) Explain why the first drops of bromine water that are added to each of the oil samples are decolourised. [2]

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- (c) The label states that the new oil contains only 15 % saturated fat. Use the results to show whether this statement is correct. [3]

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- (d) Explain why these results might persuade many consumers to use this new brand of oil in preference to the other available brands. [2]

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Examiner
only

7. Magnesium chloride is a solid with a high melting point. It conducts electricity only when molten or dissolved.

Describe the electronic changes that take place in the formation of magnesium chloride and explain its properties in terms of the bonding and structure. You may use diagrams as part of your answer. [6 QER]

END OF PAPER

6



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		



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2



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Key

A_r	Symbol	Name	Z
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Surname	Centre Number	Candidate Number
Other Names		0

**GCSE**

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S19-3430UE0-1

SCIENCE (Double Award)**Unit 5 – CHEMISTRY 2
HIGHER TIER****THURSDAY, 16 MAY 2019 – MORNING****1 hour 15 minutes**

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	15	
2.	7	
3.	6	
4.	15	
5.	11	
6.	6	
Total	60	

3430UE01
01**ADDITIONAL MATERIALS**

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Question **6** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



MAY193430UE0101

Answer all questions.

1. Crude oil can be separated into simpler mixtures, called fractions, which contain hydrocarbon compounds with boiling points in a similar range.

(a) The table lists the properties of some fractions obtained from crude oil.

Fraction	Number of carbon atoms in fraction	Boiling point range (°C)	Colour of fraction	Flame when burning	Ease of burning
fuel oil	1-4	–170 to 20	colourless	clean	very easy
petrol	5-10	20 to 70	pale yellow	clean	easy
naphtha	8-12	70 to 120	yellow	some soot	quite easy
kerosene	10-16	120 to 240	dark yellow	smoky	quite difficult
diesel oil	15-30	240 to 350	brown	smoky	difficult

Use the information in the table to describe how the burning of the fractions depends upon the size of the molecules. [2]

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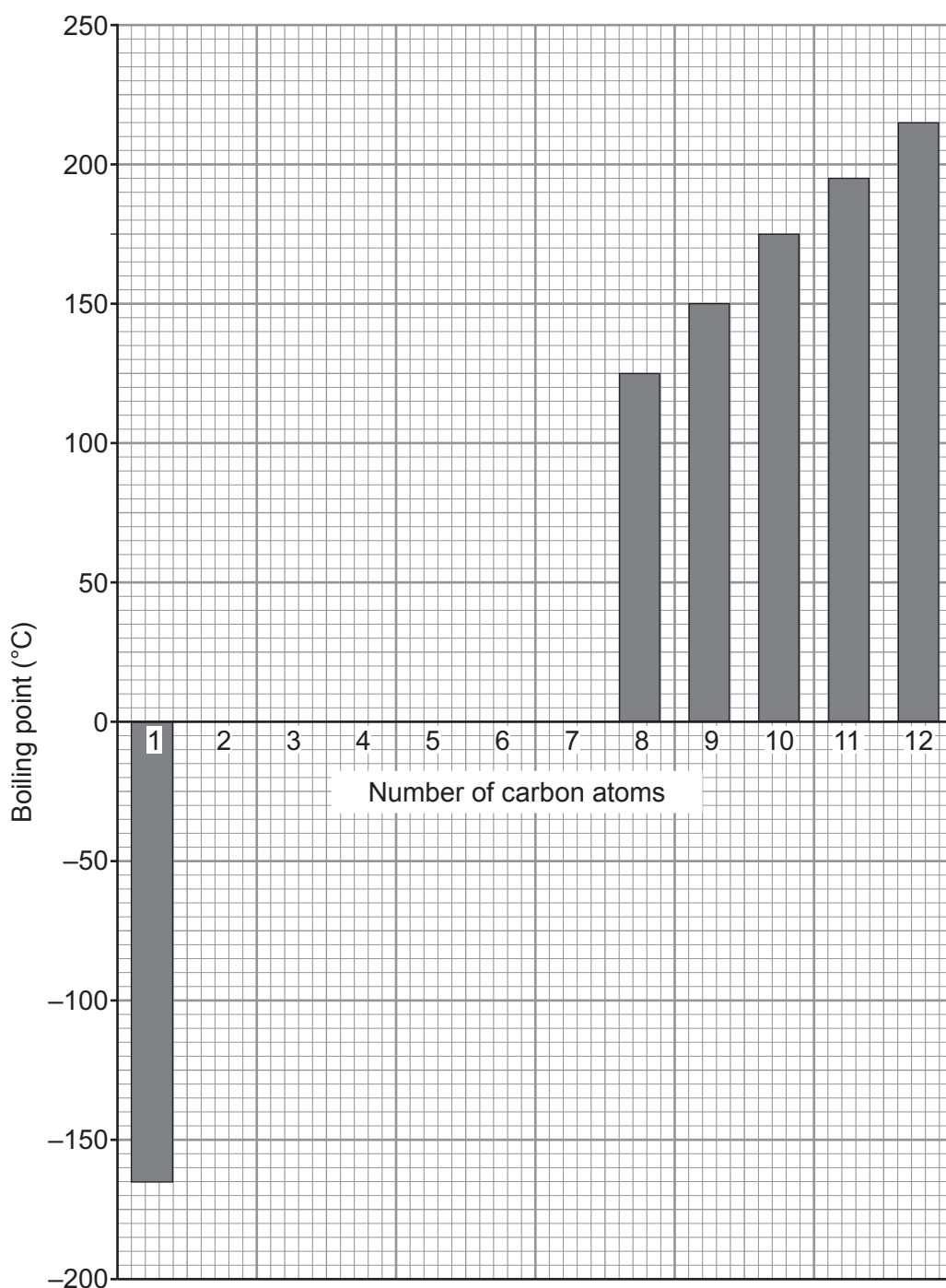
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- (b) The boiling points of hydrocarbons containing 1 to 12 carbon atoms are shown in the table below. The boiling point for the hydrocarbon with 7 carbon atoms is missing.

Number of carbon atoms	Boiling point (°C)
1	–165
2	–90
3	–40
4	10
5	35
6	70
7	
8	125
9	150
10	175
11	195
12	215



- (i) Complete the bar chart below. Some of the bars have already been drawn. [2]



- (ii) Use a ruler to draw a trend line **on the chart** and use this to estimate the boiling point of the hydrocarbon with 7 carbon atoms. [2]

Boiling point °C



(c) Many of the fractions obtained from crude oil are used as fuels.

(i) The fire triangle shows the factors necessary to start and maintain a fire.

State **one** method that could be used to safely put out a small amount of spilled petrol burning on the floor. Give the reason why your chosen method would work. [1]

Method

Reason

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(ii) One of the hydrocarbons in petrol is pentane, C_5H_{12} .

Complete and balance the symbol equation for the complete combustion of pentane. [2]



(iii) Hydrogen fuel cells are now used in many cars instead of petrol. The overall change inside a hydrogen fuel cell is the same as when hydrogen burns.

Explain why using hydrogen fuel cells in cars is better for the environment than petrol. [2]

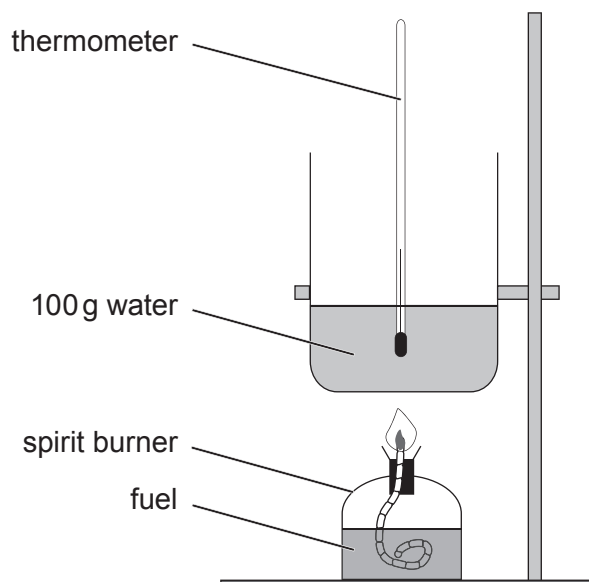
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- (d) It is possible to compare the energy released when different fuels are burned using the following apparatus.



To calculate the energy released per gram of fuel burned, the following equation is used.

$$\begin{array}{l} \text{energy released} \\ \text{per gram of fuel (J/g)} \end{array} = \frac{\text{mass of water} \times 4.2 \times \text{temperature rise (}^{\circ}\text{C)}}{\text{mass of fuel used (g)}}$$

- (i) Apart from measuring the mass of water, describe **all** the measurements that would need to be taken to be able to calculate the energy released per gram of fuel burned. [2]

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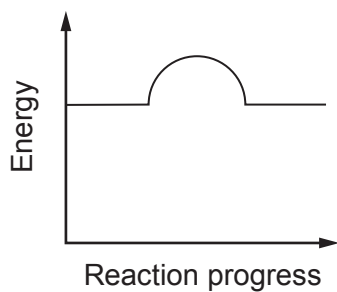
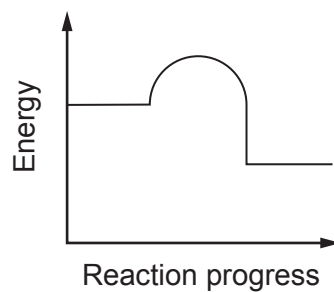
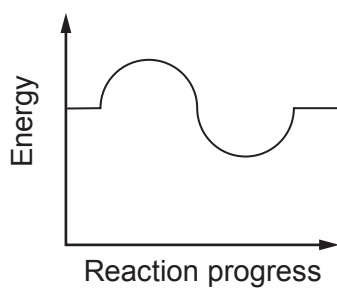
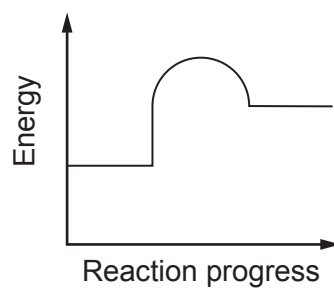
- (ii) When comparing the energy released from different fuels, 100 g of water should be used each time. State **one** other variable that should be controlled. [1]

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(iii) Tick (✓) the energy profile diagram that represents the combustion of fuels. [1]


☐

☐

☐

☐


2. (a) Magnesium reacts with oxygen to form magnesium oxide.

Using the electronic structures below, draw dot and cross diagrams to show how bonding takes place during the formation of magnesium oxide. Include the electronic structures of the ions formed. [3]

magnesium = 2,8,2

oxygen = 2,6

- (b) The melting points of magnesium oxide and sodium chloride are given below.

Substance	Melting point (°C)
sodium chloride	801
magnesium oxide	2852

Explain why there is a difference in their melting points even though they are both ionic substances. [2]

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- (c) Using the electronic structures below, draw a dot and cross diagram to show the bonding in a molecule of carbon dioxide, CO_2 . [2]

carbon = 2,4

oxygen = 2,6

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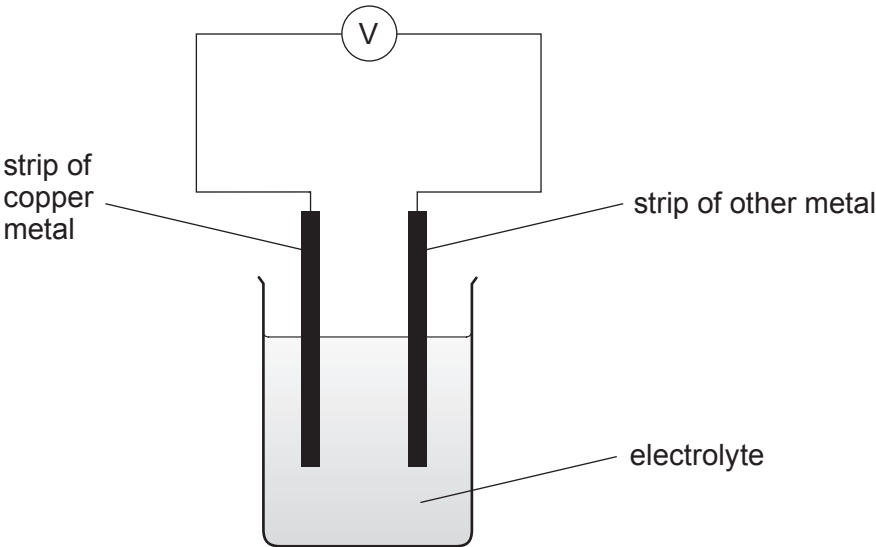
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3. When two different metals are connected in a cell, **the metal with the higher reactivity transfers its electrons to the other metal.**

The potential difference produced between pairs of metals can be used to place them in order of reactivity. **The bigger the potential difference, the bigger the difference in reactivity.**

(a) The following apparatus was used to investigate the reactivity of four different metals, **A**, **B**, **C** and **D**, compared with copper.



Each metal was placed separately into a cell with a copper strip. The potential difference was recorded for each metal and the results are shown below.

Metal	Potential difference (V)	Direction of electron flow
A	0.3	copper → metal A
B	0.6	metal B → copper
C	1.1	metal C → copper
D	0.8	copper → metal D

Use the information to place metals **A**, **B**, **C** and **D** in order of their reactivity in relation to copper. [2]

most reactive


1.
 2.
 3. copper
 4.
 5.



- (b) The reactivity of four other metals, **W**, **X**, **Y** and **Z**, was also investigated using the same apparatus. Some of the results are shown in the following table.

Pair of metals in the cell	Potential difference (V)	Direction of electron flow
W and X	1.2	W → X
W and Y	0.9	Y → W
W and Z	0.8	
Y and Z		Y → Z

The order of reactivity of these four metals is as follows.

most reactive

Y
Z
W
X

Use this information to give

- (i) the direction of the electron flow when **W** and **Z** are placed in the cell, [1]

.....

- (ii) the potential difference for the cell with metals **Y** and **Z**. [1]

..... V

- (c) When copper and zinc are placed into the cell, the following reaction takes place.



Explain how **this** reaction shows both oxidation and reduction. [2]

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4. In a class experiment, Dylan and Joel were given a solution of sodium carbonate containing 5.23 g of sodium carbonate powder dissolved in 500 cm³ of water.

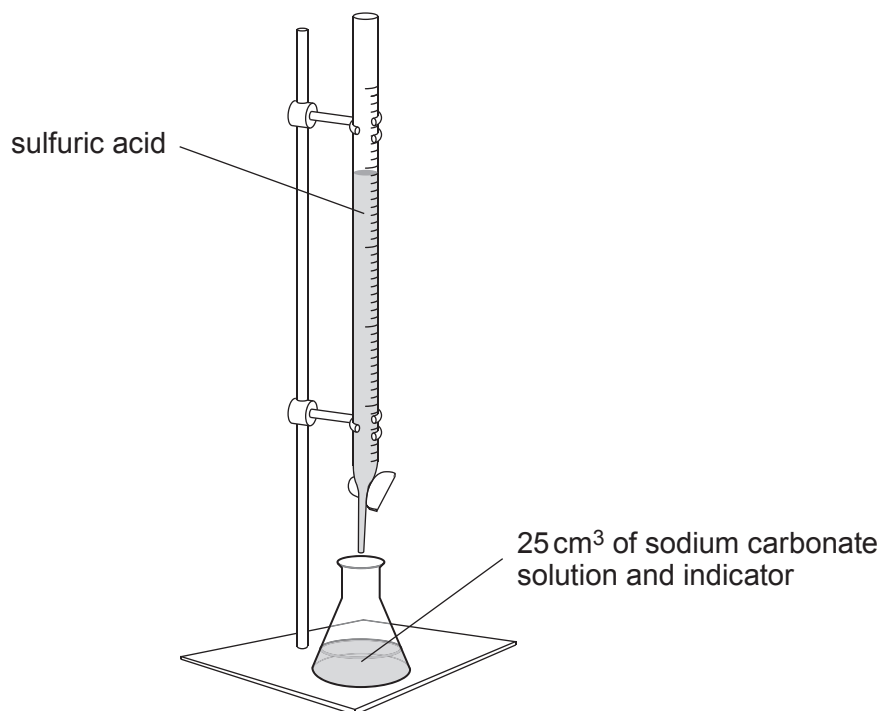
The relative formula mass (M_r) of sodium carbonate is 106.

- (a) Use this information to calculate the number of moles of sodium carbonate in the 5.23 g of the powder. Give your answer correct to **two** decimal places. [2]

Number of moles = mol

- (b) Dylan and Joel were asked to use the sodium carbonate solution to prepare a sample of sodium sulfate crystals in a three-stage process.

In the first stage of their preparation, they used the following apparatus to carry out a titration.



The equation for the reaction taking place is as follows.



- (i) A trial run was carried out and the titration repeated three times. The volume of acid added each time was recorded.

	Trial	1	2	3
Volume of sulfuric acid added (cm ³)	30.20	27.55	27.75	27.65

- I. State the purpose of carrying out a trial run. [1]

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- II. State whether the sulfuric acid or the sodium carbonate solution is the more concentrated. Give the reason for your answer. [1]

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- (ii) Use **all** of the information provided to describe **in detail** the other two stages Dylan and Joel carried out to obtain pure sodium sulfate crystals. [4]

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- (c) Sodium sulfate solution is also formed when sodium hydroxide solution reacts with copper(II) sulfate solution.

(i) Give the balanced symbol equation for this reaction. [3]

- (ii) Describe tests that can be carried out to identify both of the ions in sodium sulfate solution. Give the expected observation for both tests. [2]

- (d) The reaction occurring between solutions of sodium carbonate and magnesium sulfate forms a precipitate of magnesium carbonate.

Write the **ionic** equation for the formation of magnesium carbonate. [2]

..... + →



5. In 1985, a new allotrope of carbon was discovered and named buckminsterfullerene. This allotrope consists of sixty carbon atoms joined together to resemble a shape similar to that of a football (**Figure 1**), only ten septillion (10,000,000,000,000,000,000,000) times smaller.

This form of carbon was named after the architect Buckminster Fuller, famous for designing geodesic domes, as shown in **Figure 2**.

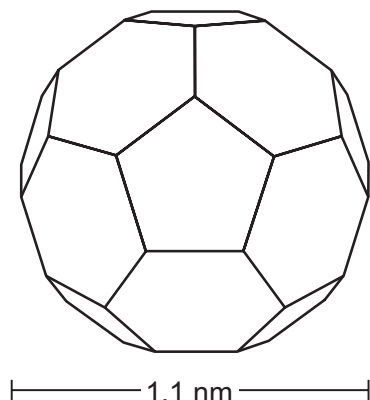


Figure 1

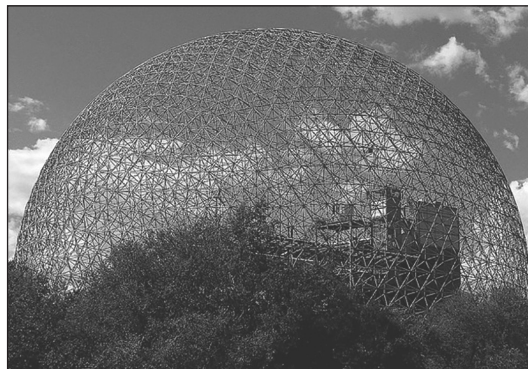


Figure 2

The structure of buckminsterfullerene is a truncated icosahedron, consisting of 20 hexagons and 12 pentagons that intersperse to form a spherical structure. Within the structure, each carbon atom is bonded to three other carbon atoms and no pentagon has a joining edge with another pentagon.

Other spherical allotropes of carbon, called fullerenes, have since been made. These include balls consisting of seventy, seventy-six and eighty-four carbon atoms. Together, they have become known as 'Buckyballs'.

Fullerenes have high melting points and boiling points. They also have a high density and a large surface area for their size.

Today, fullerenes are at the heart of nanotechnology – the study of atomic scale structures and devices. This provides many exciting new research possibilities for scientists including their potential uses in catalysts, lubricants and in nano-tubes for strengthening materials and as a way of delivering drugs into the body.

Nano-tubes are fullerenes that are used to reinforce graphite in tennis rackets because they are very strong. They are also used as semiconductors in electrical circuits.

The nano-tube's structure also allows it to be used as a container for transporting a drug in the body. A molecule of the drug can be placed inside the nano-tube cage. This keeps the drug 'wrapped up' until it reaches the site where it is needed. In this way, a dose that might be damaging to other parts of the body can be delivered safely to, for example, a tumour.



(a) Tick (✓) **all** the statements that correctly describe buckminsterfullerene.

[2]

its structure has 32 faces

☐

it has a relative molecular mass of 60

☐

it is an allotrope of carbon

☐

it has a giant ionic structure

☐

it is a hydrocarbon compound

☐

it is a smart material

☐

it is 1×10^{25} times smaller than a football

☐

(b) (i) Use the following formula to calculate the number of edges that a molecule of buckminsterfullerene has. [2]

$$\text{number of edges} = \frac{\text{total number of sides of all pentagons and hexagons}}{2}$$

Number of edges =



- (ii) Calculate, in m^3 , the approximate volume of a buckminsterfullerene molecule. Write your answer to **three** significant figures and in standard form. [3]

$$\text{volume} = \frac{4}{3} \pi r^3$$

$$\pi = 3.14$$

$$r = \text{radius}$$

$$1 \text{ nm} = 1 \times 10^{-9} \text{ m}$$

Volume = m^3

- (c) Give the **main** reason why the structure of fullerenes has resulted in there being an interest in developing their use as catalysts. [1]

.....

- (d) State why some people might oppose the use of fullerenes in drug delivery systems in the body. [1]

.....

.....

- (e) One student said that Buckyballs should be good electrical conductors but her friend disagreed.

Use your knowledge of bonding and structure to give **one** reason that **each** student could use to support their argument. [2]

.....

.....

.....

.....



Examiner
only

- 6.** Discuss what is meant by isomerism in alkanes and alkenes using compounds with the molecular formulae C_4H_{10} and C_4H_8 to illustrate your answer. [6 QER]

END OF PAPER

6



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^{-}
ammonium	NH_4^{+}	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^{-}
calcium	Ca^{2+}	fluoride	F^{-}
copper(II)	Cu^{2+}	hydroxide	OH^{-}
hydrogen	H^{+}	iodide	I^{-}
iron(II)	Fe^{2+}	nitrate	NO_3^{-}
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^{+}	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^{+}		
silver	Ag^{+}		
sodium	Na^{+}		
zinc	Zn^{2+}		





THE PERIODIC TABLE

1 2

Group

3

4

5

6

7

0

1 H Hydrogen 1																			
7 Li Lithium 3	9 Be Beryllium 4											11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10	4 He Helium 2	
23 Na Sodium 11	24 Mg Magnesium 12											27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18		
39 K Potassium 19	40 Ca Calcium 20	45 Sc Scandium 21	48 Ti Titanium 22	51 V Vanadium 23	52 Cr Chromium 24	55 Mn Manganese 25	56 Fe Iron 26	59 Co Cobalt 27	59 Ni Nickel 28	63.5 Cu Copper 29	65 Zn Zinc 30	70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36		
86 Rb Rubidium 37	88 Sr Strontium 38	89 Y Yttrium 39	91 Zr Zirconium 40	93 Nb Niobium 41	96 Mo Molybdenum 42	99 Tc Technetium 43	101 Ru Ruthenium 44	103 Rh Rhodium 45	106 Pd Palladium 46	108 Ag Silver 47	112 Cd Cadmium 48	115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54		
133 Cs Caesium 55	137 Ba Barium 56	139 La Lanthanum 57	179 Hf Hafnium 72	181 Ta Tantalum 73	184 W Tungsten 74	186 Re Rhenium 75	190 Os Osmium 76	192 Ir Iridium 77	195 Pt Platinum 78	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86		
223 Fr Francium 87	226 Ra Radium 88	227 Ac Actinium 89																	

Key

Key

A_r

Symbol

Name

Z

relative atomic mass

atomic number

Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE****3430UE0-1**

Z22-3430UE0-1

FRIDAY, 27 MAY 2022 – MORNING**SCIENCE (Double Award)****Unit 5 – CHEMISTRY 2****HIGHER TIER****1 hour 15 minutes****ADDITIONAL MATERIALS**

In addition to this examination paper you will need a calculator and a ruler.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	7	
2.	8	
3.	9	
4.	15	
5.	8	
6.	6	
7.	7	
Total	60	

3430UE01
01**INSTRUCTIONS TO CANDIDATES**

Use black ink or ball-point pen. Do not use gel pen or correction fluid. You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question 7(a) is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.



JUN223430UE0101

Answer **all** questions.

1. Polymer gels are commonly used in disposable nappies.

A company that manufactures disposable nappies was investigating the effect of temperature on the mass of water the polymer gel in their nappies is able to absorb.

- (a) The results collected using water at 40 °C are given below. The initial mass of the polymer gel bead was 0.035 g.

Time (hours)	Mass of bead (g)	Mass of water absorbed by bead (g) (to 1 decimal place)
0	0.035	0.0
2	4.048	4.0
4	6.030	6.0
6	7.280	7.2
8	7.891	7.9
10	8.181	8.1
12	8.181	8.1

- (i) The percentage increase in the mass of the bead is calculated using the following equation.

$$\text{percentage increase} = \frac{\text{mass of water absorbed}}{\text{initial mass of bead}} \times 100$$

Calculate the percentage increase in the mass of the bead after 2 hours.
Give your answer to the nearest whole number.

[1]

Percentage increase = %

- (ii) What property of polymer gels does the figure calculated in part (i) demonstrate?

[1]

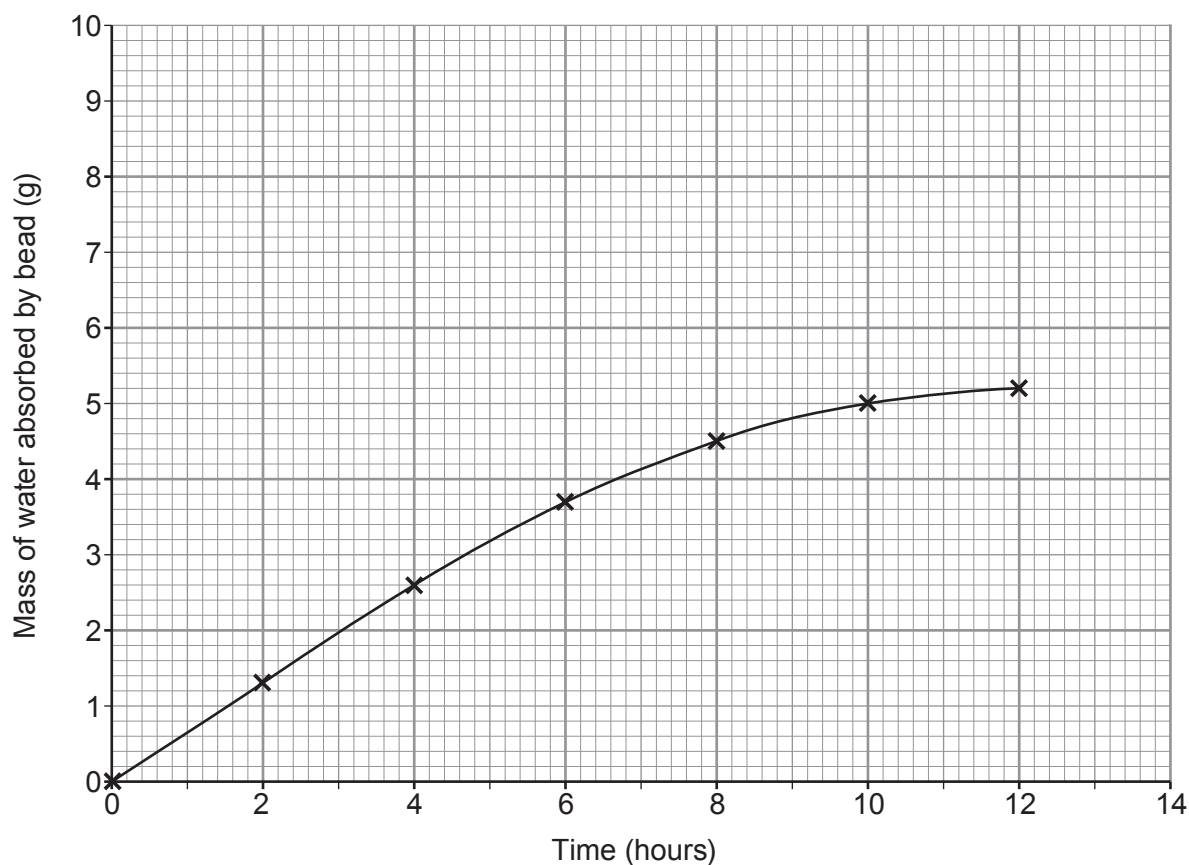
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- (b) (i) On the grid below, plot the results using water at 40 °C and draw a suitable line. Use the mass of water absorbed by the bead to 1 decimal place.

The results using water at 10 °C have already been plotted.

[3]



- (ii) Give **two** differences between the absorbing properties of the bead using water at 10 °C and at 40 °C.

[2]

Difference 1

.....

Difference 2

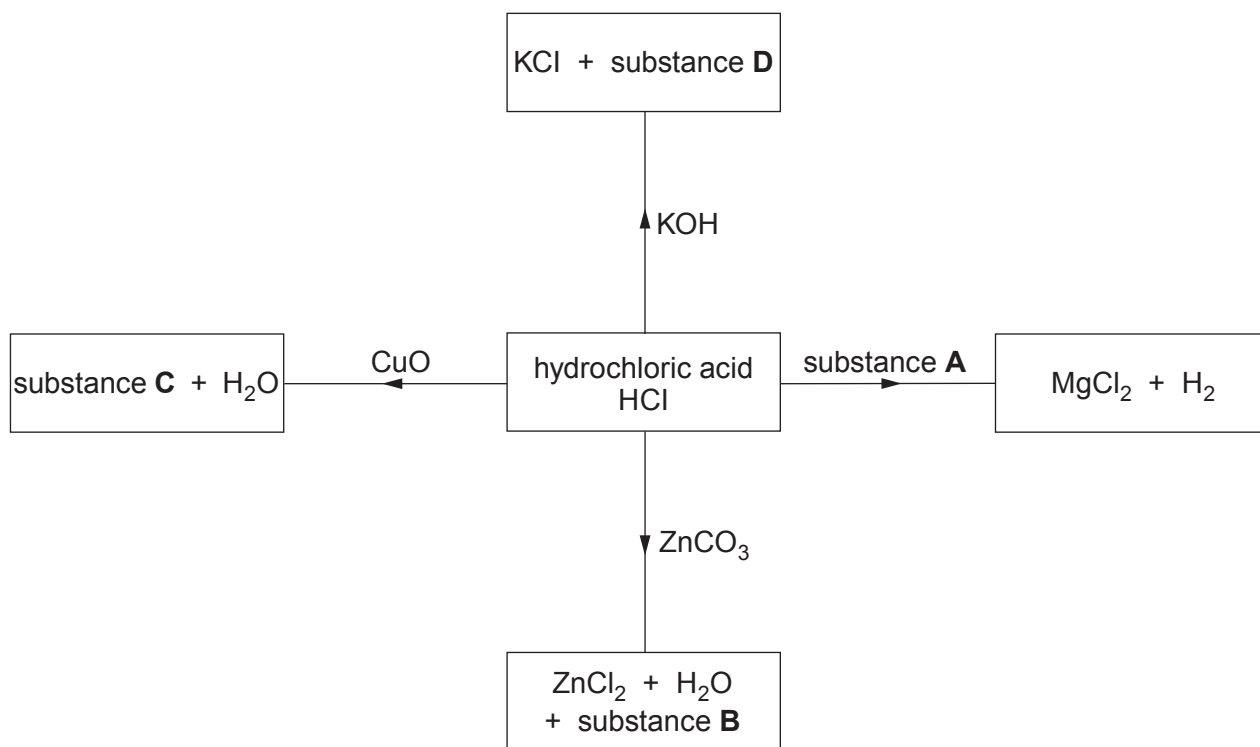
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2. The reactions of acids with metals, bases and carbonates are summarised in the following equations.

- acid + metal $\rightarrow$ salt + hydrogen
- acid + base $\rightarrow$ salt + water
- acid + carbonate $\rightarrow$ salt + water + carbon dioxide

(a) The diagram shows some reactions of hydrochloric acid, HCl.



(i) Give the **names** of substances **A** and **B**.

[2]

Substance **A**

Substance **B**

(ii) Give the **formulae** of substances **C** and **D**.

[2]

Substance **C**

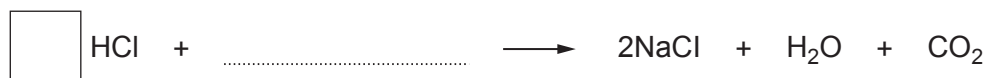
Substance **D**



- (b) Complete the equation for the reaction between hydrochloric acid and sodium carbonate by

- writing the formula of sodium carbonate on the dotted line
- putting a number into the box to balance the equation

[2]



- (c) Silver nitrate solution is used to identify the chloride ions present in hydrochloric acid.

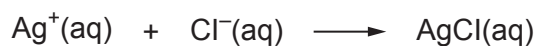
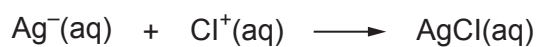
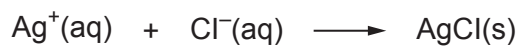
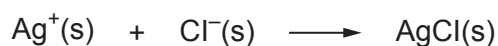
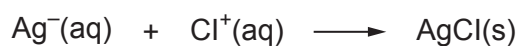
- (i) Give the observation made when silver nitrate solution is added to hydrochloric acid.

[1]

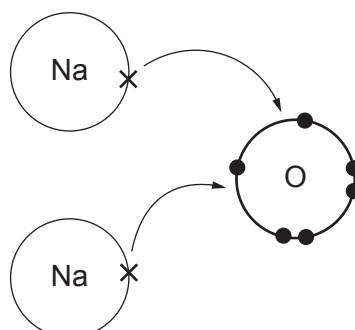
.....

- (ii) Put a tick (✓) in the box next to the correct ionic equation for the reaction between silver nitrate and hydrochloric acid.

[1]


☐

☐

☐

☐

☐


3. (a) The diagram shows the transfer of electrons that takes place during the formation of sodium oxide.



- (i) Name the type of bonding present in sodium oxide. [1]

.....

- (ii) State what must be done to sodium oxide so that it will conduct electricity.

Explain your answer.

[2]

.....

.....



- (b) (i) Draw a dot and cross diagram to show the bonding in a molecule of tetrafluoromethane, CF_4 .

[2]

carbon (C) 2,4

fluorine (F) 2,7

- (ii) Tetrafluoromethane is a simple covalent substance and is a gas at room temperature.

Explain why it has a low boiling point.

[2]

.....

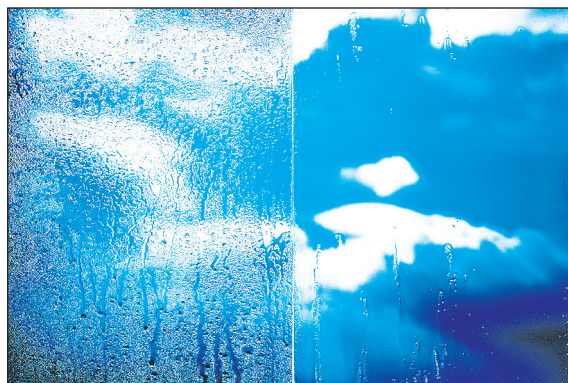
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- (c) Nanoparticles are extremely small particles that have different properties to the same material at bulk size.

Nano-scale titanium dioxide is commonly used in sunscreen and in self-cleaning windows.



Give the reason why nano-scale titanium dioxide is effective in each of these uses. [2]

Sunscreen

.....

.....

Self-cleaning windows

.....

.....



4. (a) When a mixture of iron(III) oxide and aluminium powder is heated, the following reaction takes place.



The reaction is commonly called the thermite reaction. The photographs show the reaction taking place and how it is used in the repair of railway lines.



- (i) Explain why iron is formed during the reaction.

[2]

.....

.....

.....

- (ii) Complete and balance the equation for the reaction.

[2]



- (iii) Calculate the percentage by mass of iron in iron(III) oxide, Fe_2O_3 .

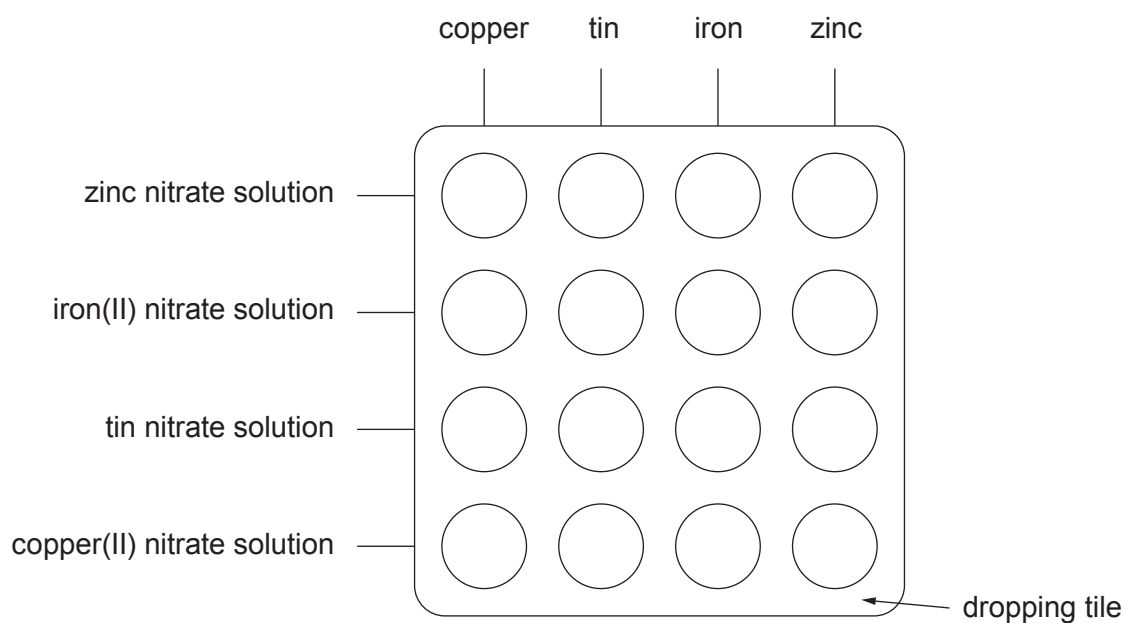
[2]

$$A_r(\text{Fe}) = 56 \qquad A_r(\text{O}) = 16$$

Percentage = %



- (b) Clare and Frankie were investigating the reactivity of metals. They carried out a series of displacement reactions in a dropping tile. In each test they placed a small piece of metal into a solution of the nitrate of a different metal as shown.



- (i) It was not necessary to carry out all of the tests. Place crosses (×) on the diagram to show which tests did not need to be carried out.

Explain your choice.

[2]

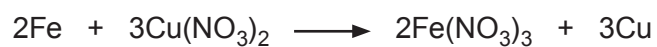
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- (ii) The equation shows the reaction between iron and copper(II) nitrate solution.



Use the equation to calculate the maximum mass of copper that you would expect to be formed when 0.224 g of iron is added to excess copper(II) nitrate solution.

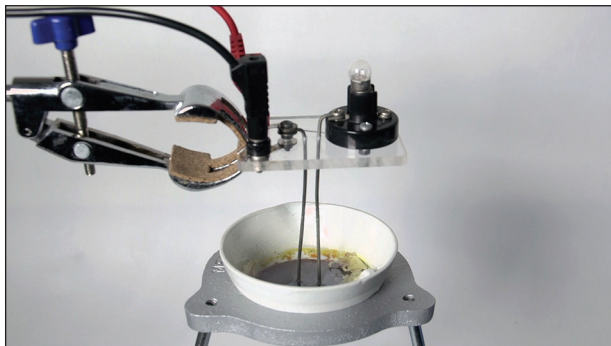
[3]

$$A_r(\text{Fe}) = 56 \quad A_r(\text{Cu}) = 63.5$$

Maximum mass of copper = g



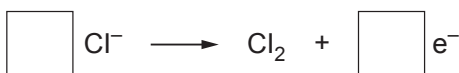
- (c) The photograph shows how the electrolysis of zinc chloride can be carried out in the laboratory.



- (i) Balance the equation that represents the reaction taking place at the anode.

Use the equation to explain the meaning of the term *oxidation*.

[2]



.....

.....

- (ii) The following method can be used to calculate the mass of zinc produced during the process.

- Record the mass of the cathode before placing it into the electrolyte
- Allow the process to run to completion
- Remove the cathode from the electrolyte and record its new mass
- Calculate the increase in mass

- I. It is often found that the increase in mass measured using this method is greater than expected. Suggest a reason for this.

[1]

.....

.....

- II. Suggest why this method cannot be used to measure the mass of chlorine produced during the process.

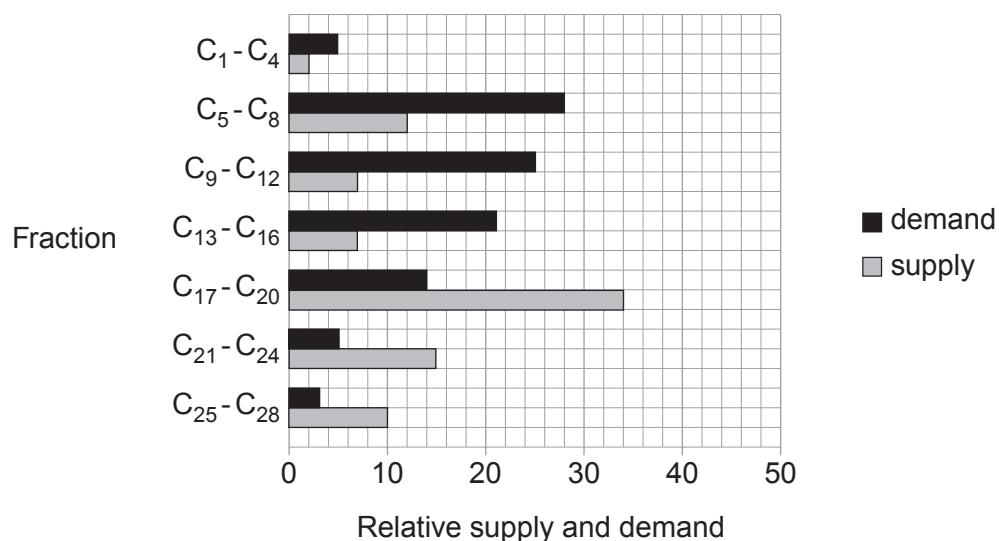
[1]

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.....



5. (a) The bar chart shows the relative supply and demand for some fractions obtained from crude oil.



- (i) Use the chart to describe how the **difference** between supply and demand of the fractions changes as the chain length increases. [2]

.....

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- (ii) The process of cracking is used to overcome the problem of insufficient supply of some fractions.

I. State what is meant by *cracking* and give the conditions needed for the process. [2]

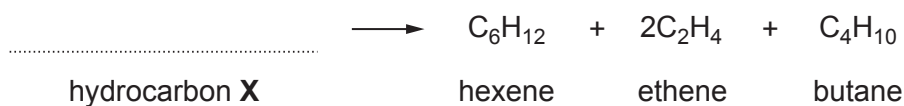
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II. When hydrocarbon **X** is cracked, it forms hexene, ethene and butane.

Give the molecular formula of hydrocarbon **X** to complete the equation. [1]



- (b) C_4H_{10} has two isomers.

(i) Give the meaning of the term *isomers*. [1]

.....

.....

.....

(ii) Draw the structures of both isomers of C_4H_{10} . [2]



6. It is well known that plastic bags are a major source of litter. These bags can exist for a very long time, ranging from tens of years in the natural environment up to a thousand years in landfill.

The use of plastic bags is now banned in some countries. These countries commonly use bags made from paper or cotton, which do not cause the same litter problem as plastic bags.

Paper or plastic?



Most people assume that a paper bag is better for the environment, however because paper bags are almost ten times heavier than plastic bags, they produce a greater mass of waste. On the other hand, because paper bags are biodegradable they do not lead to the same problem with litter as plastic bags.

When deciding whether paper is more suitable than plastic for making shopping bags, information from life cycle assessments (LCAs) of both materials must be taken into consideration.

Some of the findings from LCAs of paper and plastic bags are shown below.

Plastic bags are toxic to aquatic environments and their production is more likely to deplete natural resources

Plastic bags use 70 % less energy in their production compared to paper bags

The processes used in paper production cause atmospheric acidification and eutrophication

The production of 1 million paper bags produces 51 tonnes of carbon dioxide, compared to 31 tonnes for plastic bags

693 kJ of energy is needed to recycle 1 kg of paper bags compared to 8 kJ for the same mass of plastic bags

4 546 litres of water are used to produce 1 000 paper bags compared to 177 litres for the same number of plastic bags

Paper bags are made from trees and plastic bags are made from ethene

Paper bags are 10 times heavier than plastic bags and therefore have more associated costs

On reading these findings it is important to consider all factors relating to sustainability when deciding on the most suitable material to make shopping bags. We should not just be thinking about what happens when we throw them away!



Examiner
only

- (a) Use the information in the passage to compare the mass of waste and volume of litter generated from the use of plastic and paper bags. [2]

.....

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.....

- (b) Use the information from the LCAs to tick (✓) whether each of the factors supports or opposes the use of paper bags in preference to plastic bags. [3]

	Supports use of paper bags	Opposes use of paper bags
Impact of waste on marine life	☐	☐
Water consumption in production	☐	☐
Energy used in production	☐	☐
Carbon footprint generated in production	☐	☐
Energy used in recycling	☐	☐
Cost of transporting waste	☐	☐

- (c) The use of which type of bag is more likely to be linked to the erosion of limestone buildings?
Give a reason for your answer. [1]

Type of bag

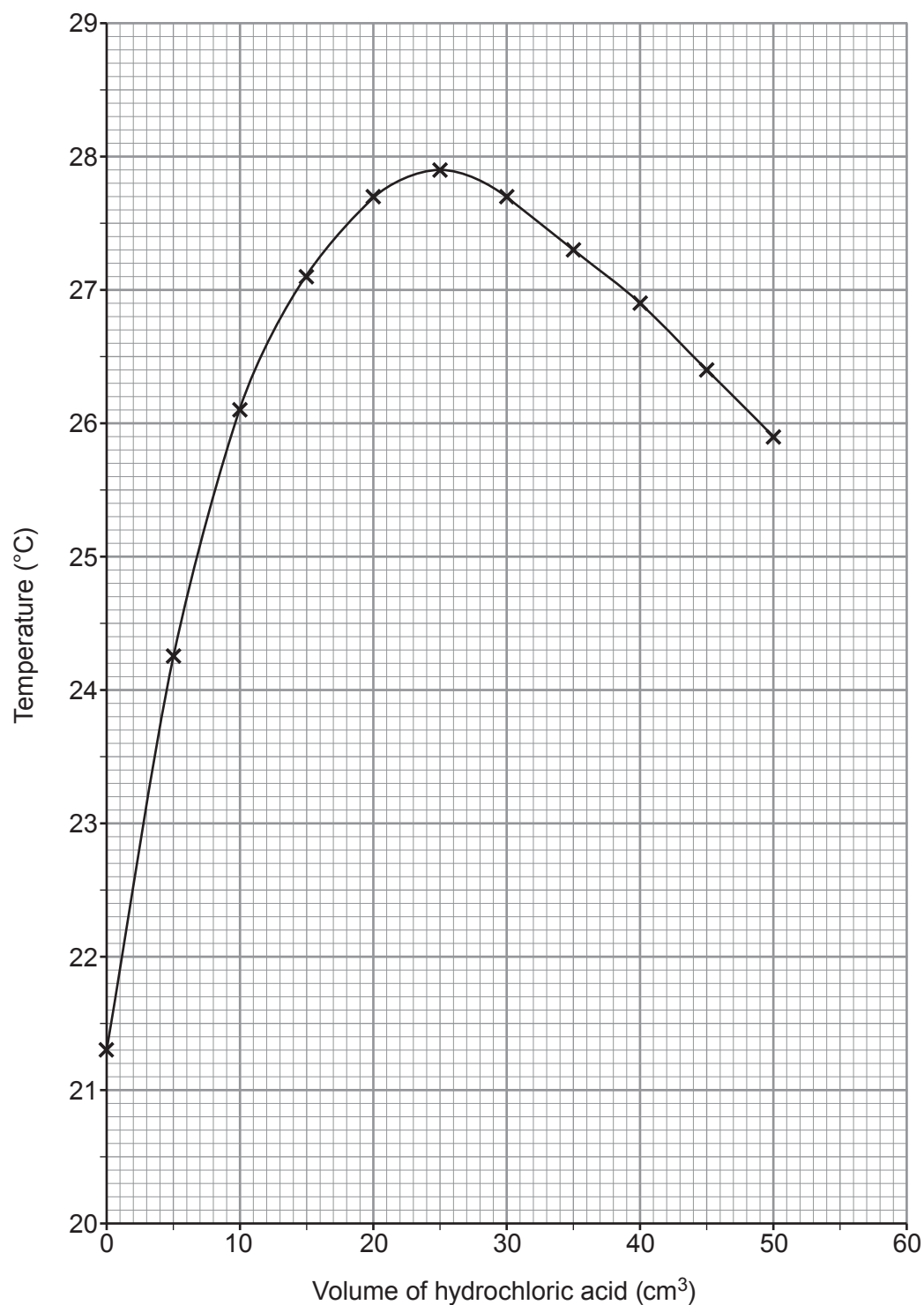
Reason

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6



7. (a) The graph shows how the temperature changes when dilute hydrochloric acid is added gradually to sodium hydroxide solution.



The pH of the solution also changes as the hydrochloric acid is added to the sodium hydroxide.

Explain the temperature changes shown on the graph and relate these to the change in pH during the reaction. You do **not** need to include equations in your answer. [6 QER]

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- (b) The reaction between hydrochloric acid and sodium hydroxide produces sodium chloride and water.

Give the **ionic** equation to show how water is formed during this reaction. [1]

.....

END OF PAPER

7



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



0

Key

A_r	Symbol	Name	Z
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Surname	Centre Number	Candidate Number
First name(s)		0

**GCSE****3430UE0-1**

S23-3430UE0-1

MONDAY, 22 MAY 2023 – MORNING**SCIENCE (Double Award)****Unit 5 – CHEMISTRY 2****HIGHER TIER****1 hour 15 minutes****ADDITIONAL MATERIALS**

In addition to this examination paper you will need a calculator and a ruler.

INSTRUCTIONS TO CANDIDATES

Use black ink or black ball-point pen. Do not use gel pen or correction fluid.

You may use a pencil for graphs and diagrams only.

Write your name, centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided in this booklet. If you run out of space, use the additional page at the back of the booklet, taking care to number the question(s) correctly.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets at the end of each question or part-question.

Question **7** is a quality of extended response (QER) question where your writing skills will be assessed.

The Periodic Table is printed on the back cover of this paper and the formulae for some common ions on the inside of the back cover.

For Examiner's use only		
Question	Maximum Mark	Mark Awarded
1.	15	
2.	7	
3.	7	
4.	9	
5.	10	
6.	6	
7.	6	
Total	60	

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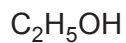
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Answer **all** questions.

1. (a) The molecular formulae of five carbon compounds, **A**, **B**, **C**, **D** and **E** are shown below.



A



B



C



D



E

- (i) Alkanes are compounds with the general formula $\text{C}_n\text{H}_{2n+2}$.

Give the **letters** of the **two** compounds that are alkanes.

[1]

..... and

- (ii) Give the **name** for compound **A**.

[1]

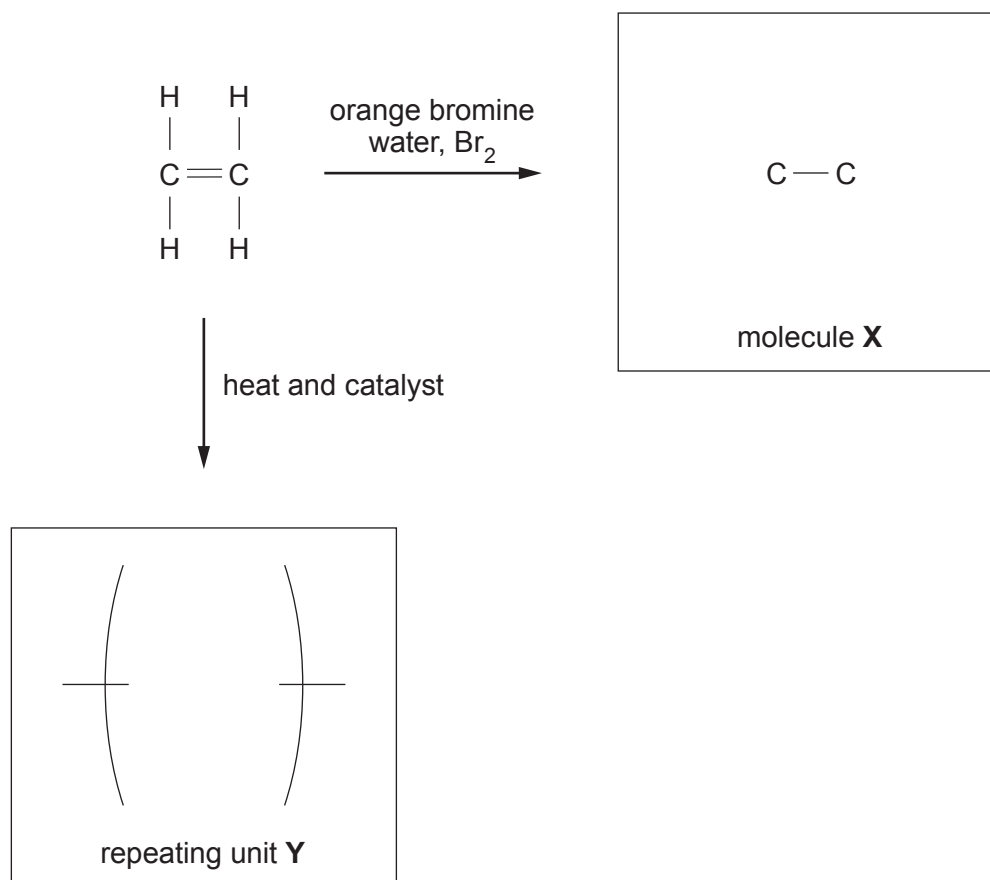
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- (iii) Draw the structural formula of compound **E**.

[1]



(b) The following diagram shows two reactions of ethene.



(i) **Complete the structural formulae** for molecule **X** and repeating unit **Y**. [2]

(ii) Describe the change that would be seen in the orange bromine water when molecule **X** is formed. [1]

(iii) Name the type of reaction that takes place to form repeating unit **Y**. [1]



(c) Paper and plastic are both used to make disposable shopping bags.



The table compares the percentages of disposable paper and plastic bags used by consumers in a Welsh town between 1988 and 2018.

Type of bag	Percentage of disposable bags used (%)						
	1988	1993	1998	2003	2008	2013	2018
paper	73	63	31	12	8	13	21
plastic	27	37	69	88	92	87	79

(i) In 2003, the consumers of the town used a total of 1.25 million disposable bags.

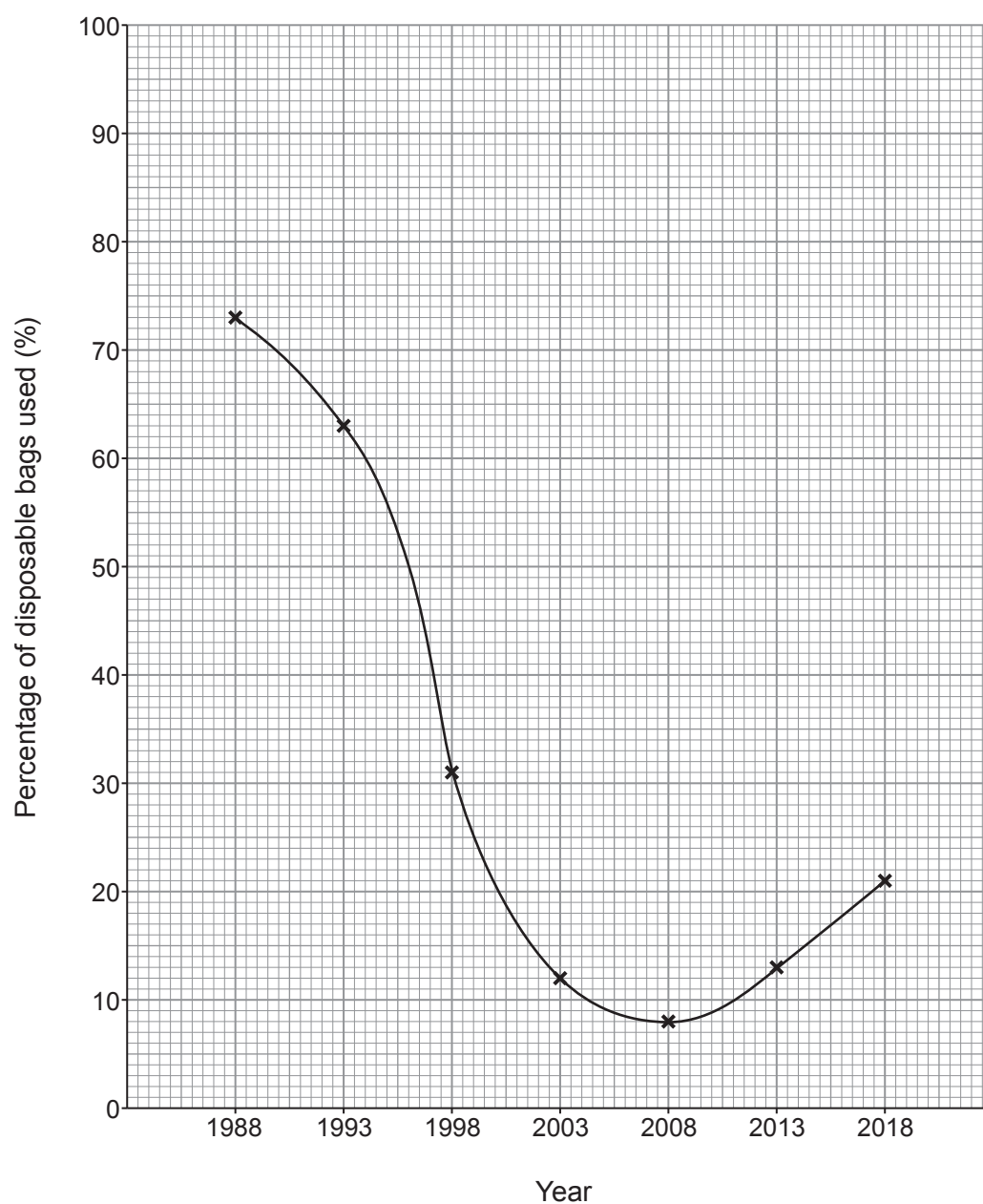
Use the information in the table to calculate the number of **paper** bags used. [2]

Number of paper bags = million



- (ii) The graph below shows the percentage of **paper** bags used between 1988 and 2018.

Plot on the **same grid** the percentage of **plastic** bags used between 1988 and 2018. Draw a suitable line. [3]



(iii) **Use the graph** to answer the following questions.

- I. Give the year when **equal** numbers of plastic bags and paper bags were used. [1]

Year

- II. Circle the **simplest** ratio that compares the use of paper bags to plastic bags in 2000. [1]

2 : 8

1 : 4

4 : 1

8 : 2

20 : 80

- (iv) Give **one** reason that explains the change seen in plastic bag usage in Wales between 2008 and 2018. [1]

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.....

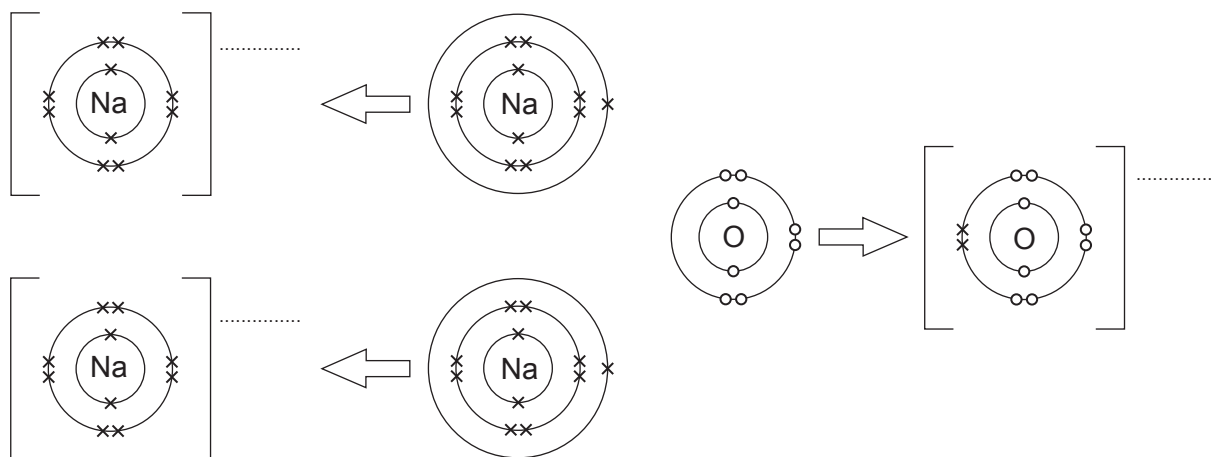


2. (a) The table gives the electronic structures of sodium and oxygen.

Element	Electronic structure
sodium	2,8,1
oxygen	2,6

- (i) Sodium reacts with oxygen to give sodium oxide.

The diagram below can be used to show the electronic changes that occur as sodium oxide is formed.



Complete the diagram by

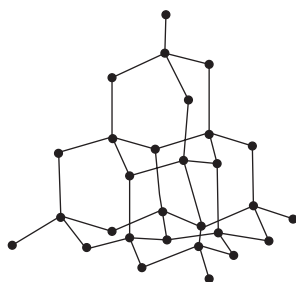
- **drawing arrows** to show the movement of electrons between the sodium and oxygen atoms,
- giving the **charges** of the sodium and oxide ions that are formed.

[2]

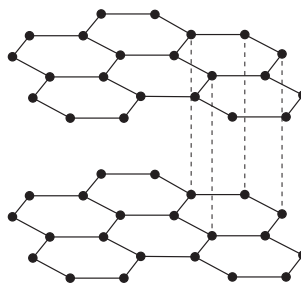


- (ii) Draw a dot and cross diagram to show the bonding in an oxygen molecule, O_2 . [2]

- (b) (i) Diamond and graphite are both made from carbon atoms. Their structures are shown below.



diamond



graphite

- I. State which structure is able to conduct electricity. Give a reason for your answer. [1]

.....

.....

- II. State which structure is used as a lubricant. Give a reason for your answer. [1]

.....

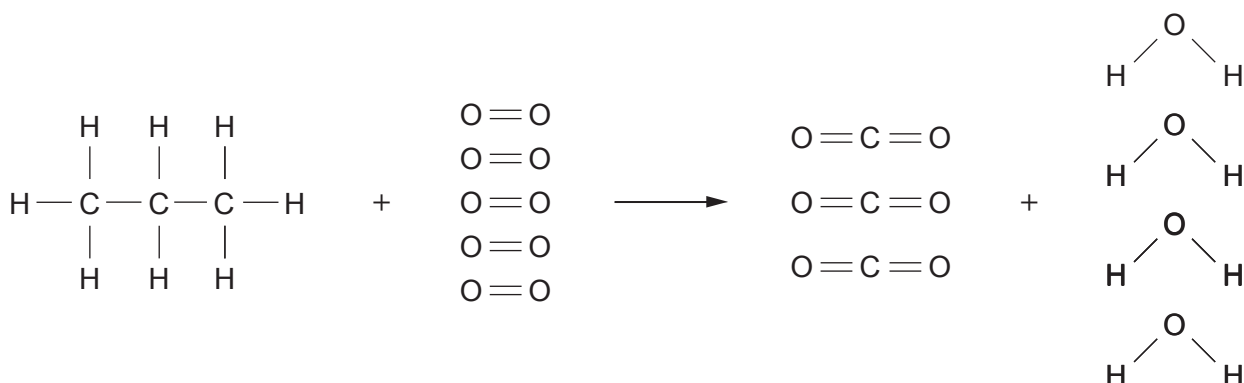
.....

- (ii) Give the name of another form of carbon that is used in drug delivery systems. [1]

.....



3. (a) Propane gas is commonly used as a fuel. The burning of propane gas is represented by the following equation.



The relative amounts of energy needed to break some of the bonds are given in the table.

Bond	Bond energy (kJ)
C—H	413
C—C	348

- (i) The total energy needed to break all of the bonds in the reactants is 6490 kJ.

Calculate the energy needed to break **one** O = O bond.

[3]

Energy needed = kJ



(ii) The energy released when all of the bonds in the products are formed is 8 542 kJ.

Tick (✓) the correct overall energy change for this reaction. [1]

2 052 kJ taken in

☐

15 032 kJ taken in

☐

15 032 kJ given out

☐

2 052 kJ given out

☐

- (b) Propane gas is commonly used in camping stoves.



The photograph shows a camping stove being used to heat 750 g of water.

When heated for 10 minutes, 220 kJ of energy was transferred to the water.

Use the following equation to calculate the temperature rise of the water.

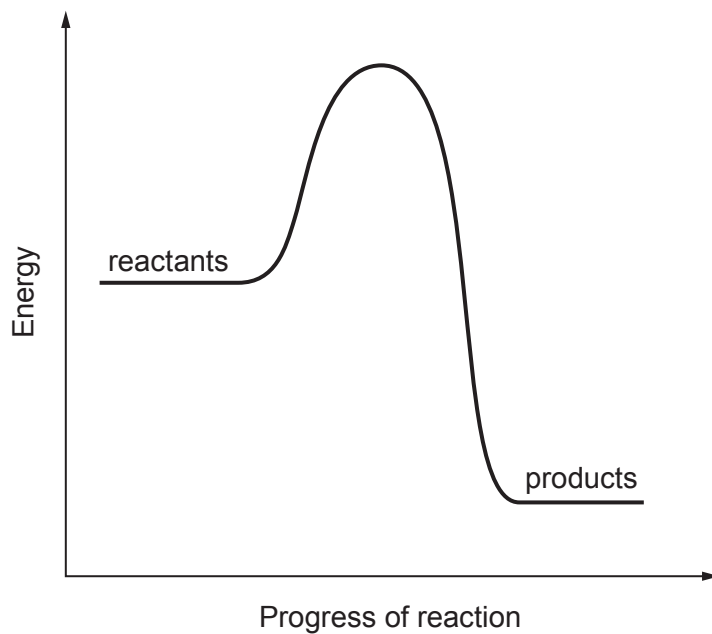
[1]

$$\text{temperature rise (}^{\circ}\text{C)} = \frac{\text{energy transferred (J)}}{4.2 \times \text{mass of water (g)}}$$

Temperature rise = $^{\circ}\text{C}$



(c) The energy profile diagram for the combustion of propane is given below.



- (i) Draw an **arrow** ($\uparrow$) **on the diagram** to show the activation energy.

Label this **A**.

[1]

- (ii) Draw an **arrow** ($\uparrow$) **on the diagram** to show the overall energy change.

Label this **B**.

[1]



4. Iron is extracted from iron oxide inside the blast furnace.



Iron oxide, coke and limestone are all fed in at the top of the furnace.

- (a) Describe the **two** main functions of coke in the furnace.

[3]

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.....

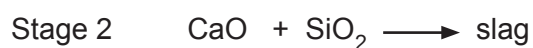
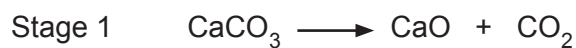
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- (b) Limestone is added to the furnace to remove some of the impurities as slag. Their removal involves a two-stage process as shown in the following equations.



- (i) A group of students were each asked to calculate the maximum mass of calcium oxide that could be produced from 8.0 g of calcium carbonate in stage 1.

The answers they calculated were as follows:

Student	Answer
Yena	4.4 g
Charlie	4.5 g
Tomasz	5.0 g

Give the name of the student that has correctly calculated the mass of calcium oxide to **one** decimal place.

You **must** show your working.

[3]

$$M_r(\text{CaCO}_3) = 100$$

$$M_r(\text{CaO}) = 56$$

Student



- (ii) In stage 2, the slag that is formed contains calcium silicate.

Calcium silicate is formed when calcium oxide (CaO) combines with silicon dioxide (SiO₂) in a 1:1 ratio, with no other product formed.

Give the chemical formula of the calcium silicate formed in stage 2.

[1]

.....

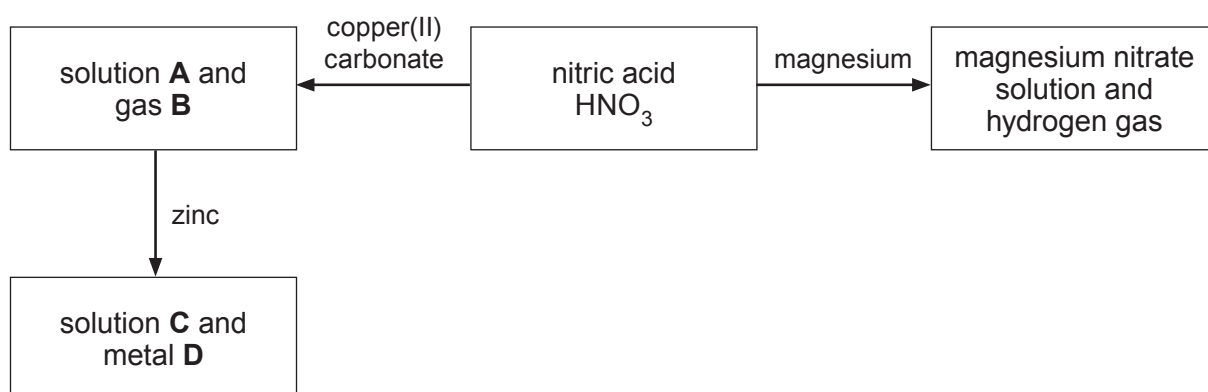
- (c) Iron reacts with oxygen to form iron(III) oxide.

Complete and balance the equation for this reaction.

[2]



5. (a) The diagram shows a sequence of reactions from nitric acid.



- (i) Use the information in the diagram to name the following substances. [3]

solution **A**

gas **B**

solution **C**

metal **D**

- (ii) Write the balanced symbol equation for the reaction between magnesium and nitric acid. [3]

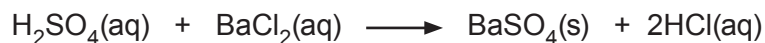
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- (iii) Name the type of reaction that takes place between solution **A** and zinc. [1]

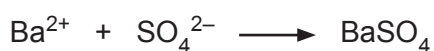
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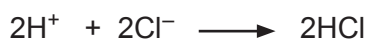
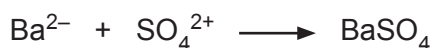


- (b) (i) The sulfate ions in sulfuric acid are identified using barium chloride solution. The equation for the reaction is shown below.



Tick (✓) the correct ionic equation for the formation of the precipitate. [1]


☐

☐

☐

☐

☐

- (ii) A student was asked to identify the ions present in barium chloride solution.

- I. Other than adding sulfate ions, state how the student would test for barium ions. Give the expected observation for a positive test. [1]

.....

.....

- II. State how the student would test for chloride ions. Give the expected observation for a positive test. [1]

.....

.....



6. Hydrogels are macromolecules, meaning they contain very large numbers of atoms.

They are polymers made from hydrophilic polymer chains. This means that they readily mix with water.

Cross-links between the polymer chains enable hydrogels to form their three-dimensional shape.

The cross-links that join the individual polymer chains together are formed by small reactive molecules that contain one or more double bonds.



The interesting feature of hydrogels is their ability to swell and shrink, depending on their surroundings. When in contact with water, hydrogels are able to absorb water and swell to many times their original size and weight, whilst maintaining the same shape.

Being able to absorb water like this has enabled hydrogels to be used in a wide range of applications. For many of these applications it is important to understand how the properties of hydrogels change with temperature.

Table 1 and **Table 2** show how temperature affects the properties of hydrogel beads in water over time.

Time (hours)	Mass of bead (g)		
	10 °C	20 °C	40 °C
0	0.035	0.035	0.035
2	1.305	2.980	4.280
4	2.512	3.681	5.904
6	3.613	4.389	6.760
8	4.298	5.195	8.101
10	5.120	5.572	8.101
12	5.400	5.980	8.101

Table 1 – the effect of temperature on the mass of hydrogel beads

Time (hours)	Diameter of bead (mm)		
	10 °C	20 °C	40 °C
0	3.56	3.56	3.56
2	10.30	13.00	18.10
4	13.40	16.50	21.70
6	16.20	18.50	23.60
8	17.80	19.60	25.24
10	19.50	20.80	25.24
12	20.50	22.00	25.24

Table 2 – the effect of temperature on the diameter of hydrogel beads



- (a) Tick (✓) the statement that **best** describes the molecules that form the cross-links within hydrogels. [1]

they are macromolecules

☐

they are unsaturated

☐

they contain hydrophilic groups

☐

they are polymers

☐

- (b) Give the phrase in the passage that shows the change the hydrogel undergoes is a physical rather than a chemical change. [1]

.....

.....

- (c) Calculate the percentage increase in **mass** of a hydrogel bead when placed in water at 20 °C for 6 hours. [2]

Percentage increase = %

- (d) Write a conclusion for the effect of **temperature** on the diameter of hydrogel beads in water over time. [2]

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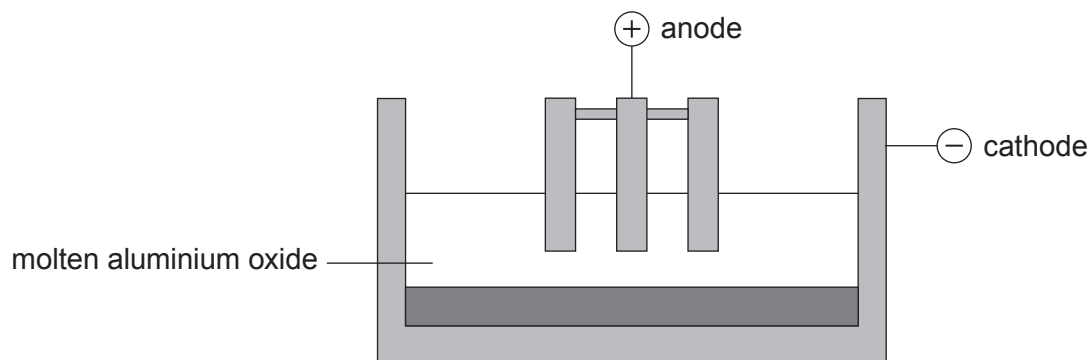
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6



7. Aluminium is extracted from its ore by electrolysis. The diagram shows a cross-section of the cell used for the extraction.



The overall equation for the extraction taking place inside the cell is given below.



Explain the movement of Al^{3+} and O^{2-} ions inside the cell and how the reactions at each electrode demonstrate oxidation and reduction. Include electrode equations in your answer.

[6 QER]

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6

END OF PAPER



FORMULAE FOR SOME COMMON IONS

POSITIVE IONS		NEGATIVE IONS	
Name	Formula	Name	Formula
aluminium	Al^{3+}	bromide	Br^-
ammonium	NH_4^+	carbonate	CO_3^{2-}
barium	Ba^{2+}	chloride	Cl^-
calcium	Ca^{2+}	fluoride	F^-
copper(II)	Cu^{2+}	hydroxide	OH^-
hydrogen	H^+	iodide	I^-
iron(II)	Fe^{2+}	nitrate	NO_3^-
iron(III)	Fe^{3+}	oxide	O^{2-}
lithium	Li^+	sulfate	SO_4^{2-}
magnesium	Mg^{2+}		
nickel	Ni^{2+}		
potassium	K^+		
silver	Ag^+		
sodium	Na^+		
zinc	Zn^{2+}		



THE PERIODIC TABLE

Group 1 2 3 4 5 6 7 0



1 28

1

H

Hydrogen

1

7 Li Lithium 3	9 Be Beryllium 4
23 Na Sodium 11	24 Mg Magnesium 12
39 K Potassium 19	40 Ca Calcium 20
86 Rb Rubidium 37	88 Sr Strontium 38
133 Cs Caesium 55	137 Ba Barium 56
223 Fr Francium 87	226 Ra Radium 88

11 B Boron 5	12 C Carbon 6	14 N Nitrogen 7	16 O Oxygen 8	19 F Fluorine 9	20 Ne Neon 10
27 Al Aluminium 13	28 Si Silicon 14	31 P Phosphorus 15	32 S Sulfur 16	35.5 Cl Chlorine 17	40 Ar Argon 18
70 Ga Gallium 31	73 Ge Germanium 32	75 As Arsenic 33	79 Se Selenium 34	80 Br Bromine 35	84 Kr Krypton 36
115 In Indium 49	119 Sn Tin 50	122 Sb Antimony 51	128 Te Tellurium 52	127 I Iodine 53	131 Xe Xenon 54
204 Tl Thallium 81	207 Pb Lead 82	209 Bi Bismuth 83	210 Po Polonium 84	210 At Astatine 85	222 Rn Radon 86

59 Co Cobalt 27	56 Fe Iron 26	55 Mn Manganese 25	52 Cr Chromium 24	51 V Vanadium 23	48 Ti Titanium 22	45 Sc Scandium 21
103 Rh Rhodium 45	101 Ru Ruthenium 44	99 Tc Technetium 43	96 Mo Molybdenum 42	93 Nb Niobium 41	91 Zr Zirconium 40	89 Y Yttrium 39
192 Ir Iridium 77	190 Os Osmium 76	186 Re Rhenium 75	184 W Tungsten 74	181 Ta Tantalum 73	179 Hf Hafnium 72	139 La Lanthanum 57
197 Au Gold 79	195 Pt Platinum 78	197 Au Gold 79	197 Au Gold 79	201 Hg Mercury 80	204 Tl Thallium 81	207 Pb Lead 82
112 Cd Cadmium 48	108 Ag Silver 47	106 Pd Palladium 46	103 Rh Rhodium 45	101 Ru Ruthenium 44	99 Tc Technetium 43	96 Mo Molybdenum 42
65 Zn Zinc 30	63.5 Cu Copper 29	59 Ni Nickel 28	56 Fe Iron 26	55 Mn Manganese 25	52 Cr Chromium 24	51 V Vanadium 23

Key

Ar

Symbol

Name

Z

relative atomic mass

atomic number